MIL-S-15701D
31 December 1985
SUPERSEDING
MIL-S-15701C
27 December 1968

MILITARY SPECIFICATION

# STITCHER, BOOK AND PAMPHLET, SINGLE STITCHING HEAD, SADDLE AND FLAT TABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This document covers an electric motor driven, single head, wire stitcher for side and saddle stitching on books or pamphlets (see 6.1).

1.2 Classification. The stitcher shall be of 7/8-inch capacity.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| J-C-580 | - Cord, Flexible, and Wire, Fixture, (Electrical, 0- to 600-Volt Service) |
| W-C-596 | - Connector, Electrical, Power, General Specification For |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3610

AMSC N/A

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| | |
|---|---|
| CC-M-1807 | - Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller) |
| QQ-W-414 | - Wire, Steel, Carbon: Bookbinder |

MILITARY

| | |
|---|---|
| MIL-D-3684 | - Printing, Duplicating, and Bookbinding Equipment, Packaging Of |
| MIL-L-11848 | - Lights, Power Tool |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification marking of US Military Property |
| MIL-STD-461 | Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference |
| MIL-STD-462 | Electromagnetic Interference Characteristics. Measurement Of |

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Publication No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)



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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When a first article is specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Standard product. The stitchers covered by this document shall be the regular commercial product of the manufacturer, except for changes necessary to comply with document requirements. All stitchers of the same size furnished under any one contract or purchase order, including parts and assemblies thereof, shall be identical.

3.3 Materials. Materials shall be of the grade and quality normally used in the manufacturer's commercial product, except where otherwise specified herein. Recycled material may be used (see 6.4).

3.4 Design. The stitchers shall be floor-mounted machines designed for side and saddle stitching of books or pamphlets by means of an electric motor driven, single stitching head. The stitching head shall automatically feed spool wire, cut the wire to a predetermined length for the thickness of work, form a staple, and imbed the formed staple in books and pamphlets. The imbedded staple shall be clinched by the action of a rising clincher.

3.5 Construction.

3.5.1 Capacities and dimensions. The stitchers shall have capacities and dimensions as listed in table I.



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TABLE I. Capacity and dimensions

| | Size 7/8 |
|---|---|
| Floor space, maximum | 25 by 28 inches |
| Net weight, maximum | 290 pounds |
| Capacity: | |
| Stitching thickness, range | 2 sheets to 7/8 inch |
| Throat depth, minimum | |
| Side stitching | 8 inches |
| Saddle stitching | 14 inches |
| Wire size: | |
| Round gage | Nos. 24 through 28 |
| Flat | No. 20 by 24 |
| | No. 20 by 25 |
| | No. 21 by 25 |
| Wire draw, adjustable, range | 1 1/16 inches to 2 13/16 inches |
| Crown (width of stitch) | 9/16 inch |
| Speed (nominal) | 125 stitches per minute |
| Clincher height, from floor | 34 inches ± 1 inch |
| Table, flat, minimum | 5 1/2 inches by 26 inches |
| Number of stitching heads | 1 |

3.5.2 Supporting structure. The supporting structure shall be a one or two piece casting designed for floor mounting with floor mounting holes and shall support the entire assembly in such a manner to meet performance requirements specified herein.

3.5.3 Electric motor. Each stitcher shall be powered by a constant speed electric motor of 1725 revolutions per minute (rpm) conforming to the applicable requirements of CC-M-1807, for operation on the power source specified (see 6.2).

3.5.3.1 Motor mounting. The electric motor shall be mounted on a supporting bracket that is in turn mounted on the supporting structure. The motor supporting bracket shall be provided with elongated holes for horizontal and vertical alignment of the electric motor and clutch drive.



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3.5.3.2 Electrical controls. The electric motor shall be controlled by means of a reset type, thermal overload, "off-on" toggle switch having a thermal heater compatible with the electric motor. The switch shall be mounted on the supporting structure at a location accessible to the operator at, or above, the clincher height.

3.5.4 Work light. A work light conforming to type I or II of MIL-L-11848 shall be furnished with each stitcher, except that the unit shall be mounted and wired directly to the stitcher and 2-conductor cable will be permitted in lieu of 3-conductor cable.

3.5.5 Table. The work table shall be designed for side and saddle stitching and shall be retained in each position by means of hand operable clamps or bolts. Tilting of the work table to either position shall not require readjustment of the clincher or stitching head elevation.

3.5.5.1 Position guides. Position guides shall be furnished with the work table for use in side stitching operations and shall be removable for saddle stitching. The guide shall consist of one member (backguide) running at least 75 percent of the table length, with elongated mounting holes less than 1 inch long at right angles to its length, for adjustment purposes. Sufficient clearance shall be provided in this guide to preclude the stitcher striking the guide at any setting. In addition, sliding left and right hand guides shall be furnished with the backguide. Each sliding guide shall incorporate a hand operable clamping device for locking the sliding guide at any location along the backguide.

3.5.6 Clutch. The power shall be transmitted from the electric motor to the stitching head and clincher by means of a mechanical clutch. The clutch shall be actuated by a foot treadle mounted for operator accessibility from a sitting or standing position. When pressure is released from the foot treadle, the stitching head and clincher shall return to the starting position.

3.5.7 Thickness adjustment. A hand operated lever or wheel shall be incorporated in the machine for operator adjustment of length of wire draw, as required, for varying thickness of work within the range specified in table I. A positive stop shall be incorporated to preclude possible jamming of the stitching head against the work table. An independent hand-operable compression adjustment is permissible.



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3.5.8 Power cord and plug. The stitchers shall be equipped with not less than 7 nor more than 8 feet of type S, SO, ST, SJO, or SJT flexible electrical cord conforming to J-C-580. The cord shall terminate in a series D, grounding type, male, plug connector conforming to W-C-596.

3.5.9 Electric wiring. The stitchers shall be furnished fully wired in accordance with the applicable provisions of the NFPA publication No. 70 - National Electrical Code.

3.5.10 Electromagnetic compatibility. When specified (see 6.2), the end item shall be designed and equipped for electromagnetic compatibility in accordance with class II B of MIL-STD-461 (see 4.6.1).

3.5.11 Lubrication. All moving parts requiring lubrication shall be equipped with the appropriate lubrication fitting for the type of lubricant required. The end item shall be serviced with grades of lubricants designed for use in the temperature range of 50°F to 125°F. After lubrication, the equipment shall be tagged in a conspicuous place to indicate the temperature range and regular commercial designation of the lubricants used.

3.5.12 Safety. All rotating parts accessible to personnel shall be shielded by removable guards for operator safety.

3.5.13 Adjusting devices. All parts requiring adjustment to compensate for varying working conditions shall be provided with readily accessible adjusting devices.

3.6 Performance characteristics. The machines, without bolting to the floor, shall not creep on the supporting surface when stapling material under conditions outlined in 4.6.2 through 4.6.2.1.1. All staples shall be clinched with the staple legs of equal length and either turned toward the stapled material, or parallel to and against the stapled material. There shall be no evidence of oil stains or other soil marks on stapled material. The formed staples shall not bind in the stitching head or clincher, nor cause any malfunction of the machine (see 4.6.2.1).

3.6.1 Shipboard requirements, Navy. When specified for Naval shipboard use (see 6.2 and 6.5), the entire unit shall withstand, without failure, the stress and shock resulting from the listing and rolling of vessels through 45 degrees and sudden motion of the supporting deck, both vertically and horizontally. In addition, the unit shall be operable at a list or roll of 10 degrees.



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3.7 Marking for identification. Marking for identification shall be in accordance with unit marking requirements of MIL-STD-130.

3.8 Equipment manual. The contractor shall furnish an equipment manual with each stitcher. The manual contents and additional distribution shall be as specified in the contract or purchase order (see 6.2).

3.9 Finish. The finish shall be in accordance with the manufacturer's commercial practice. Painted surfaces shall be uniformly coated and free from runs, teardrops, grit, and sags.

3.10 Workmanship. The end item shall be clean and free of corrosion, burrs, snags, rough or sharp edges, and projections hazardous to using personnel. All components and assemblies shall be in proper alignment with all fastening devices drawn tight. Welded and soldered connections, when applicable, shall show complete fusion and shall be free of undercut, burn through, and breaks in and adjacent to the welded and soldered areas. All functioning components shall operate smoothly without binding.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When specified (see 3.1 and 6.2), one sample stitcher shall be subjected to the examinations in 4.5.1.1 and 4.5.1.2 and the tests specified in 4.6. Failure of any examination or test shall constitute failure of the first article inspection.

4.4 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.



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4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 End item examination. The lot shall consist of all stitchers offered for inspection at one time. The sample unit shall be one complete, fully assembled stitcher.

* 4.5.1.1 Visual examination. Sample stitchers shall be selected at random from each lot and shall be examined for defects as classified in table II. The inspection level shall be II, and the acceptable quality level (AQL) shall be 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Finish | Not manufacturer's commercial finish | | 201 |
| | Painted surfaces not uniformly coated or not free from runs, teardrops, grit, and sags | | 202 |
| Construction and workmanship | Any part fractured, split, punctured, dented, bowed, malformed, or otherwise impaired | 101 | |
| | Sharp burr, sliver, splinter, or hazardous projection | 102 | |
| | Component or assembly improperly aligned | 103 | |
| | Welded or soldered connection, when applicable, improperly fused, burned through, undercut, or breaks in or adjacent to weld area | 104 | |
| | Fastening device not tight | | 203 |
| | Fastening device missing | 105 | |
| | Functioning component binds | 106 | |



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TABLE II. Classification of defects - Continued

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Construction and workmanship (continued) | | | |
| Safety guards | Removable guards not provided around rotating parts for operator safety | 107 | |
| Supporting structure | Not provided with floor mounting holes | 108 | |
| | Elongated holes not provided in motor support bracket for alignment of motor | 109 | |
| | Work light not mounted and wired to stitcher | | 204 |
| Table | Work table not designed for side and saddle stitching or not retained by hand-operable clamps or bolts | 110 | |
| | Position guides not furnished or not adjustable | 111 | |
| | Sliding guides not furnished with hand-operable clamps | | 205 |
| Electrical | Wiring cut, abraded, not properly joined, or loose at terminal | 112 | |
| | Adequate slack not provided to relieve strain | 113 | |
| | Excessive insulation stripped from wiring | 114 | |



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TABLE II. Classification of defects - Continued

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Construction and workmanship (continued) | | | |
| Lubrication | Any lubrication fitting missing where required | | 206 |
| | Not lubricated | 115 | |
| | Not tagged to indicate commercial lubricant required | | 207 |
| Manuals | Missing, illegible, or not as specified | | 208 |
| Marking for identification | Missing, incomplete, not legible, or not affixed in prescribed manner | 116 | |

4.5.1.2 Dimensional and weight examination. Sample stitchers shall be selected at random from each lot and shall be examined for dimensions and weights as specified in table I. Any deviation shall constitute a defect. The inspection level shall be S-2, and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.5.1.3 End item testing. When a first article is not required, the initial production unit shall be tested in accordance with 4.6.1, when applicable. In addition, production units shall be tested as specified in 4.6.2 through 4.6.2.1.1 using inspection level S-2, and the AQL shall be 6.5, expressed in terms of defects per hundred units.

4.5.2 Packaging inspection. Examination of preservation packing, and marking shall be in accordance with section 5 and MIL-D-3684.



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4.6 Tests.

4.6.1 Electromagnetic compatibility test. When electromagnetic compatibility is required, the preproduction sample or initial unit of production shall be tested by the contractor in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or his private testing organization. The contractor shall furnish to the contracting officer for approval the interference control plan, the EMI/EMC test plan, and the electromagnetic compatibility test report required in MIL-STD-461. Upon approval of the test report by the contracting officer and provided all other requirements of this document are met, the preproduction sample or initial unit of production, as applicable, shall be used as a model for all production units.

4.6.2 Performance. Machines shall be set up for operation in accordance with the manufacturer's instructions without bolting to floor surface. Book paper and bookbinder wire conforming to QQ-W-414, in each of the applicable sizes indicated in table I, shall be used for the stitching tests. Throughout all of the following tests, observation shall be made to determine that all controls and adjustments function as intended, that the machine does not creep on the floor, and that the quality of stitching conforms to 3.6.

4.6.2.1 Operational. Each applicable wire size in table I shall be threaded into the machine in turn. The machine shall be run continuously for * not less than 1/2 minute for each wire size, while stapling 2 sheets of offset book paper. The formed staples of each size shall be counted and calculations shall be made to determine compliance with the applicable speed requirement in table I.

4.6.2.1.1 Stitching. Not less than 50 staples of each applicable wire size shall be set in two (2) sheets of offset book paper. The test shall then be repeated using offset book paper piled to 50 percent of capacity and then to 100 percent of capacity to determine compliance with applicable requirements of table I. One half of each wire size shall be side stitched 1/4 inch from the paper edge using the position guides. The other half shall be saddle stitched.

## * 5. PACKAGING

5.1 Preservation and packing.

5.1.1 Levels A and B. Preservation and packing shall be in accordance with the applicable level A or B requirements of MIL-D-3684 as specified (see 6.2).



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5.1.2 Commercial. Preservation and packing shall be in accordance with ASTM D 3951.

5.2 Marking. In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The stitcher is intended for use in printing and reproduction shops for stitching two or more pages of pamphlets or book sections together.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1).
c. Power source electrical characteristics required (see 3.5.3).
d. When electromagnetic compatibility is required (see 3.5.10).
e. When equipment is required for Naval shipboard use (see 3.6.1 and 6.6).
f. Manual contents and distribution (see 3.8).
g. Applicable levels of preservation and packing (see 5.1).

* 6.3 First article. When a first article inspection is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for inspection and approval of first article test results, and disposition of first articles.

* 6.4 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.3).

6.5 Navy use. When stitchers are for shipboard use, the using activity should furnish appropriate tests and procedures to determine compliance with 3.6.1.

6.6 Electromagnetic compatibility. The contracting officer should submit a copy of the interference control plan, the EMI/EMC test plan, and the test report required (see 4.6.1) to the Commanding General, U.S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey 07703, for approval.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-S-15701D | STITCHER, BOOK AND PAMPHLET, SINGLE STITCHING HEAD, SADDLE AND FLAT TABLE |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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